

MLIBQ Analytics - Getting Started

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MLIBQ Quant Analytics

- Available for use in London, New York, Tokyo, Hong Kong ...

Features

- Multi-Yield Curves with CSA Support
- Bond Curves, Bond Pricing and Risk
- Swaps Pricing and Risk
- Credit Derivatives
- Rates Derivatives

MLIBQ Analytics - Getting Started

Excel Launcher

\\fs002\Global\Secure\DIR_MLib_Analytics\MLIB_Excel_Addin\LAUNCH_MLIBQ_CURRENT_UAT.bat
\\fs002\Global\Secure\DIR_MLib_Analytics\MLIB_Excel_Addin\LAUNCH_MLIBQ_CURRENT_UAT_X64.bat

32-Bit

64-Bit

Features

Function Groups for Ease of Navigation: meDate, meCurve, meSwap ...
Multi-Curve Construction
Swap Pricing & Risk
Bond Curve Stripping, Pricing & Risk
Credit Modelling, Pricing & Risk
Rates Derivatives Pricing & Risk

Documentation

\\fs002\Global\Secure\DIR_MLib_Analytics\Documentation

Excel Examples

\\fs002\Global\Secure\DIR_MLib_Analytics\Examples

Workshops

\\fs002\Global\Secure\DIR_MLib_Analytics\Workshop

- Intra-day & On-Demand Library Updates
- Many releases a day, takes ca. 15mins

MLIB Dojo

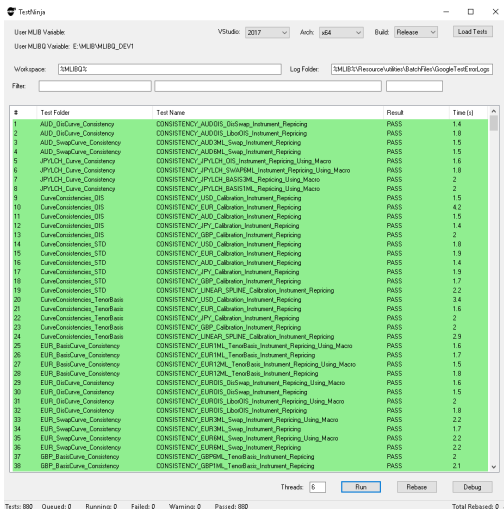
Read MLIB Variable Load Default Config Save Default Config Save My Local Config

☐ User MLIB Variable
☒ User MLIBQ Variable: E:\MLIB\MLIBQ_DEV1

	Description	Batch File Path	実行!
1	TESTNINJA	%MLIBQ%\resource\utils\BatchFiles\TestNinja\TestNinja.exe	Run 1
2	SET %MLIBQ% to MLIBQ_DEV1	E:\MLIB\MLIBQ_DEV1\resource\utils\BatchFiles\GIT\SetMibEnvironment\variables\Set_MLIBQ_Er	Run 2
3	SET %MLIBQ% to MLIBQ_DEV2	E:\MLIB\MLIBQ_DEV2\resource\utils\BatchFiles\GIT\SetMibEnvironment\variables\Set_MLIBQ_Er	Run 3
4	SET %MLIBQ% to MLIBQ_DEV3	E:\MLIB\MLIBQ_DEV3\resource\utils\BatchFiles\GIT\SetMibEnvironment\variables\Set_MLIBQ_Er	Run 4
5	SET %MLIBQ% to MLIBQ_DEV4	E:\MLIB\MLIBQ_DEV4\resource\utils\BatchFiles\GIT\SetMibEnvironment\variables\Set_MLIBQ_Er	Run 5
6	SET %MLIBQ% to MLIBQ_DEV5	E:\MLIB\MLIBQ_DEV5\resource\utils\BatchFiles\GIT\SetMibEnvironment\variables\Set_MLIBQ_Er	Run 6
7	GOODZILLA: Renew Licence	%MLIBQ%\resource\utils\BatchFiles\GIT\Godzilla\MLIB\RenewExcelLicenceMLIBQ.bat	Run 7
8	Deploy SERVER API:: LONDON	%MLIBQ%\resource\utils\BatchFiles\GIT\BuildServer\BUILD_SERVER_DEPLOY_MLIB_API_32_A	Run 8
9	LOCAL OVERNIGHT BUILD	%MLIBQ%\resource\utils\BatchFiles\GIT\UpdateBuild\lites\l4.bat	Run 9
10	Deploy EXCEL LDN X86	%MLIBQ%\resource\utils\BatchFiles\GIT\DeployAddn\LONDON_DeployMLIBQ_ADDIN_PROD.ba	Run 10
11	Deploy EXCEL TOK X86	%MLIBQ%\resource\utils\BatchFiles\GIT\DeployAddn\TOKYO_DeployMLIBQ_ADDIN_PROD.bat	Run 11
12	Deploy EXCEL LDN X86 and X64	%MLIBQ%\resource\utils\BatchFiles\GIT\DeployAddn\LONDON_DeployMLIBQ_ADDIN_PROD_X	Run 12
13	Deploy EXCEL TOK X86 and X64	%MLIBQ%\resource\utils\BatchFiles\GIT\DeployAddn\TOKYO_DeployMLIBQ_ADDIN_PROD_X86	Run 13
14			Run 14
15			Run 15

Recent Commands: -

- Every release is tested:
- 880 Test Scenarios and > 30,000 individual tests



The screenshot shows the TestMingui application interface. At the top, there are settings for 'User MUB Variable' (E:\MUB\MUB0_DEV1), 'VStudio' (2017), 'Arch' (x64), 'Build' (Release), and a 'Load Tests' button. Below these are fields for 'Workspace' (\\MUB02) and 'Log Folder' (\\MUB02\resources\mlbq\BatchFiles\GoogleTest\ErrorLog). A 'Filter' field is also present. The main area is a table with columns: #, Test Folder, Test Name, Result, and Time (s). The table lists 38 test scenarios, all of which have a 'PASS' result. The bottom status bar shows 'Tests: 880', 'Queued: 0', 'Running: 0', 'Failed: 0', 'Warning: 0', 'Passed: 880', and 'Total Released: 0'. There are also buttons for 'Threads: 6', 'Run', 'Release', and 'Debug'.

#	Test Folder	Test Name	Result	Time (s)
1	AUD_OisCurve_Conistency	CONSISTENCY_AUDOIS_OisSwap_Instrument_Repricing	PASS	1.4
2	AUD_OisCurve_Conistency	CONSISTENCY_AUDOIS_LboOIS_Instrument_Repricing	PASS	1.8
3	AUD_SwapCurve_Conistency	CONSISTENCY_AUD3M_Swap_Instrument_Repricing	PASS	1.5
4	AUD_SwapCurve_Conistency	CONSISTENCY_AUD3M_Swap_Instrument_Repricing	PASS	1.5
5	JPLCH_Curve_Conistency	CONSISTENCY_JPLCH_OIS_Instrument_Repricing_Using_Macro	PASS	1.6
6	JPLCH_Curve_Conistency	CONSISTENCY_JPLCH_SwapBML_Instrument_Repricing_Using_Macro	PASS	1.8
7	JPLCH_Curve_Conistency	CONSISTENCY_JPLCH_BASIS3M_Repricing_Using_Macro	PASS	2
8	JPLCH_Curve_Conistency	CONSISTENCY_JPLCH_BASIS1M_Repricing_Using_Macro	PASS	1.5
9	CurveConsistencies_OIS	CONSISTENCY_USD_Calibration_Instrument_Repricing	PASS	4.2
10	CurveConsistencies_OIS	CONSISTENCY_EUR_Calibration_Instrument_Repricing	PASS	1.5
11	CurveConsistencies_OIS	CONSISTENCY_AUD_Calibration_Instrument_Repricing	PASS	1.4
12	CurveConsistencies_STO	CONSISTENCY_JPY_Calibration_Instrument_Repricing	PASS	2
13	CurveConsistencies_STO	CONSISTENCY_GBP_Calibration_Instrument_Repricing	PASS	1.8
14	CurveConsistencies_STO	CONSISTENCY_USD_Calibration_Instrument_Repricing	PASS	1.9
15	CurveConsistencies_STO	CONSISTENCY_EUR_Calibration_Instrument_Repricing	PASS	1.4
16	CurveConsistencies_STO	CONSISTENCY_JPY_Calibration_Instrument_Repricing	PASS	1.7
17	CurveConsistencies_STO	CONSISTENCY_GBP_Calibration_Instrument_Repricing	PASS	2.2
18	CurveConsistencies_STO	CONSISTENCY_USD_Calibration_Instrument_Repricing	PASS	3.4
19	CurveConsistencies_TenorBasis	CONSISTENCY_EUR_Calibration_Instrument_Repricing	PASS	2
20	CurveConsistencies_TenorBasis	CONSISTENCY_JPY_Calibration_Instrument_Repricing	PASS	2
21	CurveConsistencies_TenorBasis	CONSISTENCY_GBP_Calibration_Instrument_Repricing	PASS	2.9
22	CurveConsistencies_TenorBasis	CONSISTENCY_USD_Calibration_Instrument_Repricing	PASS	1.6
23	CurveConsistencies_TenorBasis	CONSISTENCY_EUR_Calibration_Instrument_Repricing	PASS	1.7
24	CurveConsistencies_TenorBasis	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	1.5
25	EUR_BasisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing	PASS	1.8
26	EUR_BasisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	1.6
27	EUR_BasisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing	PASS	1.5
28	EUR_OisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	1.5
29	EUR_OisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing	PASS	2
30	EUR_OisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	1.8
31	EUR_OisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing	PASS	1.5
32	EUR_OisCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	1.5
33	EUR_SwapCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing	PASS	2.2
34	EUR_SwapCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	1.7
35	EUR_SwapCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing	PASS	2.2
36	EUR_SwapCurve_Conistency	CONSISTENCY_EUR1M_TenorBasis_Instrument_Repricing_Using_Macro	PASS	2
37	GBP_BasisCurve_Conistency	CONSISTENCY_GBP1M_TenorBasis_Instrument_Repricing	PASS	2.1
38	GBP_BasisCurve_Conistency	CONSISTENCY_GBP1M_TenorBasis_Instrument_Repricing	PASS	2.1

MLIB Analytics Ribbon



Highlights

Curve Calibration

Open the Curve Calibration Toolkit

Clear MLIB Cache

Clear the MLIB Object Memory

MLIB Version

Show the MLIB Version and License Expiry Date!!!

Quick Save / Load

Save / Load MLIB Objects

MLIB Browser

Navigate to the MLIB Examples Folder

Sticky Handles

Toggle the Object Safety Feature On / Off (discussed later)

Record MLIB Functions

Record / Log MLIB Functions

Pulse Workbook

Refresh Workbook Every 5 Seconds

MLIBQ Excel Toolbar

- Examples & Pricing Tools
- Helper Functions and Methods

Function Groups

meDate...

meDateBusinessDays
meDateCentralBank
meDateECB
meDateFromTenor
meDateFuturesContract
meDateIMMCurrent
meDateIMMNext
meDateIMMnth
meDateToday
meDateYearFraction

meCurve...

meCurveCalibrateOIS
meCurveCalibrateSwap
meCurveCalibrateBasis
meCurveCalibrateFXForwards
meCurveDiscountFactors
meCurveForwardRates

meLWO...

meLWOExists
meLWOType
meLWOList
meLWODElete
meLWODEleteAll
meLWOSave
meLWOLoad

meUtilities...

meUtilityClean
meUtilityClearLWOCache
meUtilityDataFilter
meUtilityInterpolation
meUtilityLVB
meUtilityPCA
meUtilityParallelModeEnable
meUtilityResize
meUtilityRecord
meUtilityReplay
meUtilityVersion

meLWOBond...

meLWOBondCreate
meLWOBondDirtyPrice
meLWOBondCleanPrice
meLWOBondAccruedInterest
meLWOBondAccruedInterestDays
meLWOBondDisplaySchedule
meLWOBondDisplayCashflows
meLWOBondYield
meLWOBondCompoundYield
meLWOBondModifiedDuration
meLWOBondDV01
meLWOBondDV01Numerical

meLWOCurve...

meLWOCurveBumpAll
meLWOCurveBumpInstrument
meLWOCurveCalibrate
meLWOCurveDisplay
meLWOCurveForwardRatesTable
meLWOCurveMarketDataCreate

meLWOSchedule...

meLWOScheduleCreate
meLWOScheduleCreateBespoke
meLWOScheduleDisplay

meLWOSwap...

meLWOSwapCreate
meLWOSwapCreateFromGenerator
meLWOSwapAddFee
meLWOSwapDisplayCashflows
meLWOSwapDisplay
meLWOSwapAnnuity
meLWOSwapParRate
meLWOSwapParSpread
meLWOSwapPV01
meLWOSwapDV01
meLWOSwapDeltaLadder

Other Function Groups

meLWOCapFloor...
meLWOAssetSwap...
meLWOCreditDefaultSwap...
meLWOSwaption...
meLWOTotalReturnSwap...
meLWOVariableNotionalSwap...

BusinessDayAdjustment

MOD_FOLLOWING
FOLLOWING
MOD_PRECEDING
PRECEDING
NO_CHANGE

Daycounts

30/360
ACT/360
ACT/365FJ
ACT/365
ACT/365_ISDA
ACT/ACT
ACT/ACT_ISMA

Calendar Codes *(Sourced from SwapsWire)*

☐ ADB (Adelaide)	☐ HKB (Hong Kong)	☐ PAB (Paris)
☐ AEB (AbuDhabi)	☐ HKS (Hong Kong Stock Exchange trading)	☐ PEB (Perth)
☐ AMB (Amsterdam)	☐ HAB (Hanoi)	☐ PRB (Prague)
☐ ANB (Ankara)	☐ HEB (Helsinki)	☐ RJB (Rio de Janeiro)
☐ ATB (Athens)	☐ HMB (Hamilton(Bermuda))	☐ RIB (Riyadh)
☐ AUB (Auckland)	☐ HOB (Hobart)	☐ RKB (Reykjavik)
☐ BAB (Beunos Aires)	☐ ISB (Istanbul)	☐ SOM (Eurex trading (German equity contracts))
☐ BMA (US Treasuries (SIFMA recommended close))	☐ JAB (Jakarta)	☐ SON (Eurex trading (STOXX contracts))
☐ BBB (Brisbane)	☐ JOB (Johannesburg)	☐ SPB (Sao Paulo)
☐ BCB (Bucharest)	☐ KLB (Kuala Lumpur)	☐ SAB (Santiago)
☐ BDB (Budapest)	☐ LAB (Labuan)	☐ SEB (Seoul)
☐ BEB (Beijing)	☐ LBB (Lubljana)	☐ SHB (Shanghai)
☐ BKB (Bangkok)	☐ LMB (Lima)	☐ SIB (Singapore)
☐ BMB (Mumbai)	☐ LNB (London)	☐ SOB (Sofia)
☐ BOB (Bogota)	☐ LSB (Lisbon)	☐ STB (Stockholm)
☐ BRB (Brussels)	☐ LXB (Luxembourg)	☐ SXB (Stuttgart)
☐ BTB (Bratislava)	☐ MDB (Madrid)	☐ SYB (Sydney)
☐ BXB (Brasilia)	☐ MEB (Melbourne)	☐ TYO (ISDA Standard TYO)
☐ CMF (Chicago Mercantile Exch. trading (Equity))	☐ MIB (Milan)	☐ TGT (TARGET)
☐ CHS (Chicago Stock Exchange trading)	☐ MNB (Manila)	☐ TKB (Tokyo)
☐ CNB (Canberra)	☐ MOB (Montreal)	☐ TPB (Taipei)
☐ COB (Copenhagen)	☐ MWB (Moscow)	☐ TRB (Toronto)
☐ DEL (Delaware)	☐ MXB (Mexico City)	☐ VIB (Vienna)
☐ DBB (Dubai)	☐ NBR (Brazilian payments system)	☐ WAB (Warsaw)
☐ DRB (Darwin)	☐ NYB (New York)	☐ WEB (Wellington)
☐ DUB (Dublin)	☐ NYS (New York Stock Exchange trading)	☐ ZAB (Zagreb)
☐ FRB (Frankfurt)	☐ NDQ (NASDAQ)	☐ ZUB (Zurich)
☐ GEB (Geneva)	☐ OSB (Oslo)	

Date Functions

meDate...

[meDateBusinessDays](#)
[meDateCentralBank](#)
[meDateECB](#)
[meDateFromTenor](#)
[meDateFromYearFraction](#)
[meDateFuturesContract](#)
[meDateIMMFromMonth](#)
[meDateMMFromMonth](#)
[meDateIsHoliday](#)
[meDateIsWeekday](#)
[meDateIsWeekend](#)
[meDateIsWorkingDay](#)
[meDateNthIMM](#)
[meDateNthIMMFromStartDate](#)
[meDateYearFraction](#)

meDateBusinessDays

From	Mon, 2-Sep-19
To	Mon, 9-Sep-19
Calendar	TGT
NumberOfBusinessDays	5

meDateFromTenor

StartDate	Mon, 2-Sep-19
Tenor	6M
BusinessDayAdjustment	MOD_FOLLOWING
Calendar	TGT
ResultDate	Mon, 2-Mar-20

meDateFromYearFraction

StartDate	Mon, 2-Sep-19
YearFraction	0.5
DayCount	ACT/365
ResultDate	Tue, 3-Mar-20

meDateYearFraction

StartDate	Mon, 2-Sep-19
EndDate	Tue, 5-Dec-17
DayCount	ACT/365
YearFraction	-1.742465753

meDateFuturesContract

Ticker	Z9
ResultDate	Wed, 18-Dec-19

meDateIMMFromMonth

Year	2017
Month	3
Calendar	NYB
BusinessDayAdjustment	MOD_FOLLOWING
ResultDate	Wed, 15-Mar-17

meDateCentralBank

CentralBankID	ECB
ReferenceDate	Mon, 2-Sep-19
NextCentralBankDate	Thu, 12-Sep-19

meDateECB

ReferenceDate	Mon, 2-Sep-19
NextECBDate	Thu, 12-Sep-19

meDateIsHoliday

ReferenceDate	Mon, 2-Sep-19
Calendar	LN8:NYB:TKB
Result	TRUE

meDateIsWeekend

ReferenceDate	Mon, 2-Sep-19
Result	FALSE

meDateIsWeekday

ReferenceDate	Mon, 2-Sep-19
Result	TRUE

meDateIsWorkingDay

ReferenceDate	Mon, 2-Sep-19
Calendar	LN8:NYB:TKB
Result	FALSE

Swap Schedule Generation

meProductSwapSchedule

EffectiveDate	02-Sep-19
MaturityDate	2Y
AccrualDaycount	ACT/360
AccrualFrequency	Quarterly
AccrualBusinessDayAdjustment	MOD_FOLLOWING
AccrualCalendar	TGT
PaymentFrequency	Quarterly
PaymentBusinessDayAdjustment	MOD_FOLLOWING
PaymentCalendar	TGT
PaymentLag	0D
FixingBusinessDayAdjustment	MOD_FOLLOWING
FixingCalendar	TGT
FixingLag	0D
FixingAdvanceOrArrear	ADVANCE
FirstStubDate	
LastStubDate	
RollDay	IMM

FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate
Mon, 2-Sep-19	Mon, 2-Sep-19	Wed, 18-Sep-19	16	0.0444	Wed, 18-Sep-19
Wed, 18-Sep-19	Wed, 18-Sep-19	Wed, 18-Dec-19	91	0.2528	Wed, 18-Dec-19
Wed, 18-Dec-19	Wed, 18-Dec-19	Wed, 18-Mar-20	91	0.2528	Wed, 18-Mar-20
Wed, 18-Mar-20	Wed, 18-Mar-20	Wed, 17-Jun-20	91	0.2528	Wed, 17-Jun-20
Wed, 17-Jun-20	Wed, 17-Jun-20	Wed, 16-Sep-20	91	0.2528	Wed, 16-Sep-20
Wed, 16-Sep-20	Wed, 16-Sep-20	Wed, 16-Dec-20	91	0.2528	Wed, 16-Dec-20
Wed, 16-Dec-20	Wed, 16-Dec-20	Wed, 17-Mar-21	91	0.2528	Wed, 17-Mar-21
Wed, 17-Mar-21	Wed, 17-Mar-21	Wed, 16-Jun-21	91	0.2528	Wed, 16-Jun-21
Wed, 16-Jun-21	Wed, 16-Jun-21	Thu, 2-Sep-21	78	0.2167	Thu, 2-Sep-21

Yield Curves

- Single & Global Curves
- Risk-Free Curves (ARR Curves) in Progress
- 2 Step Process: Update Market Data then Calibrate
- Low Latency - Curve Calibration approx 10 milliseoncds

MARKETDATA PROPERTIES

Currency	USD
CurveType	OIS
CurveIndexFrequency	DAILY
AsOfDate	Mon, 2-Sep-19

Create Market Data Handle

Handle Name	USD_OIS_CURVE_MARKETDATA
Market Data Handle	USD_OIS_CURVE_MARKETDATA@1:3

Curve Dictionary

Handle Name	USD_OIS_CURVE
Curve Generator	USD_OIS
Domestic Curve Collection	USDYC
Foreign Curve Collection	

Curve Build

Handle Name	USD_OIS_CURVE@2:3
Curve Collection	USDYC
Curve Index	OIS:USDOIS:USDDF:USDDF_USDCSA:USDDF_NOCSA
Calibration Time Taken	0.01600 sec
Curve Loaded	TRUE

OIS

Term	Rate	StartDate	EndDate	Use
ON	2.12000%			TRUE
1W	2.12000%			TRUE
2W	2.12100%			TRUE
3W	2.04400%			TRUE
1M	1.99150%			TRUE
2M	1.90860%			TRUE
3M	1.84400%			TRUE
4M	1.77930%			TRUE
5M	1.73190%			TRUE
6M	1.68300%			TRUE
9M	1.56450%			TRUE

LIBOR OIS BASIS SPREADS

Term	Rate
1Y	0.29250%
2Y	0.27875%
3Y	0.27000%
4Y	0.26500%
5Y	0.26250%
7Y	0.25875%
10Y	0.25375%
12Y	0.24815%
15Y	0.24625%
20Y	0.24250%
25Y	0.24125%
30Y	0.24125%

SWAPS

Term	Rate
1Y	1.78830%
2Y	1.51150%
3Y	1.40000%
4Y	1.34200%
5Y	1.31820%
7Y	1.33000%
10Y	1.38920%
12Y	1.42810%
15Y	1.47510%
20Y	1.51950%
25Y	1.54530%
30Y	1.55070%

MARKETDATAPROPERTIES

Currency	USD
CurveType	SWAP
CurveIndexFrequency	QUARTERLY
AsOfDate	Mon, 2-Sep-19

Create Market Data Handle

Handle Name	USD_SWAP_3M_CURVE_MARKETDATA
Market Data Handle	USD_SWAP_3M_CURVE_MARKETDATA@3:2

Curve Dictionary

Handle Name	USD_SWAP_3M_CURVE
Curve Generator	USD_SWAP_3M
Domestic Curve Collection	USDYC
Foreign Curve Collection	

Curve Build

Handle Name	USD_SWAP_3M_CURVE@4:2
Curve Collection	USDYC
Curve Index	STD:USD3ML
Calibration Time Taken	0.00700 sec
Curve Loaded	TRUE

LIBORFIXINGS

Term	Rate
3M	2.1328%
6M	2.0240%

FUTURES

Term	StartDate	EndDate	Rate	Volatility
ED1 Comdty	18-Sep-19	18-Dec-19	97.938	0.700%
ED2 Comdty	18-Dec-19	18-Mar-20	98.140	0.700%
ED3 Comdty	18-Mar-20	17-Jun-20	98.450	0.700%
ED4 Comdty	17-Jun-20	16-Sep-20	98.600	0.700%
ED5 Comdty	16-Sep-20	16-Dec-20	98.705	0.700%
ED6 Comdty	16-Dec-20	17-Mar-21	98.730	0.700%
ED7 Comdty	17-Mar-21	16-Jun-21	98.825	0.700%
ED8 Comdty	16-Jun-21	15-Sep-21	98.845	0.700%

SWAPS

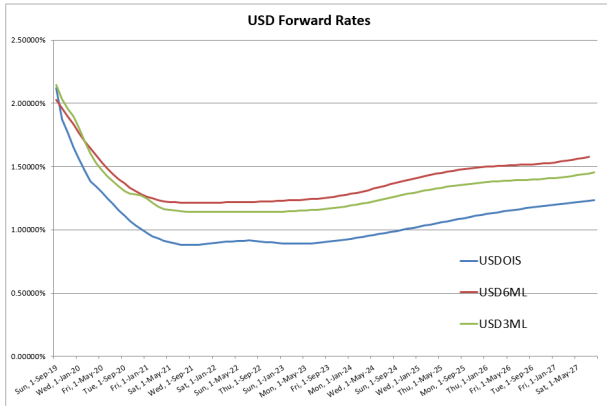
Term	Rate	Use
3Y	1.4000%	TRUE
4Y	1.3420%	TRUE
5Y	1.3182%	TRUE
7Y	1.3300%	TRUE
8Y	1.3430%	TRUE
9Y	1.3630%	TRUE
10Y	1.3892%	TRUE
12Y	1.4281%	TRUE
15Y	1.4751%	TRUE
20Y	1.5195%	TRUE
25Y	1.5453%	TRUE
30Y	1.5507%	TRUE

Forward Rates

CurveCollection	USDYC		
CurveIndices	USDOIS	USD3ML	USD6ML

meLWOCurveForwardRatesTable

Fixing Dates	USDOIS	USD3ML	USD6ML
Mon, 2-Sep-19	2.12000%	2.14279%	2.02824%
Wed, 2-Oct-19	1.87441%	2.03139%	1.96293%
Mon, 4-Nov-19	1.76571%	1.95812%	1.89243%
Mon, 2-Dec-19	1.65061%	1.89596%	1.83082%
Thu, 2-Jan-20	1.56037%	1.80831%	1.76578%
Mon, 3-Feb-20	1.46849%	1.69887%	1.69972%
Mon, 2-Mar-20	1.38088%	1.60310%	1.64278%
Thu, 2-Apr-20	1.34127%	1.52405%	1.58617%
Mon, 4-May-20	1.29566%	1.47089%	1.53036%
Tue, 2-Jun-20	1.25003%	1.42271%	1.48499%
Thu, 2-Jul-20	1.20161%	1.38053%	1.43945%
Mon, 3-Aug-20	1.15266%	1.34318%	1.39854%
Wed, 2-Sep-20	1.10934%	1.30816%	1.36476%
Fri, 2-Oct-20	1.06812%	1.28717%	1.33327%
Mon, 2-Nov-20	1.03263%	1.27815%	1.30553%
Wed, 2-Dec-20	1.00157%	1.26943%	1.28268%
Mon, 4-Jan-21	0.97198%	1.24510%	1.26187%
Tue, 2-Feb-21	0.94942%	1.21422%	1.24727%
Tue, 2-Mar-21	0.93071%	1.18440%	1.23565%
Fri, 2-Apr-21	0.91301%	1.16472%	1.22644%
Mon, 3-May-21	0.89985%	1.15725%	1.22027%
Wed, 2-Jun-21	0.89009%	1.15001%	1.21652%
Fri, 2-Jul-21	0.88346%	1.14560%	1.21455%
Mon, 2-Aug-21	0.88016%	1.14372%	1.21377%
Thu, 2-Sep-21	0.88005%	1.14184%	1.21372%
Mon, 4-Oct-21	0.88253%	1.14137%	1.21406%
Tue, 2-Nov-21	0.88575%	1.14130%	1.21441%
Thu, 2-Dec-21	0.89003%	1.14125%	1.21491%
Mon, 3-Jan-22	0.89521%	1.14120%	1.21550%
Wed, 2-Feb-22	0.90000%	1.14114%	1.21607%
Wed, 2-Mar-22	0.90441%	1.14105%	1.21678%

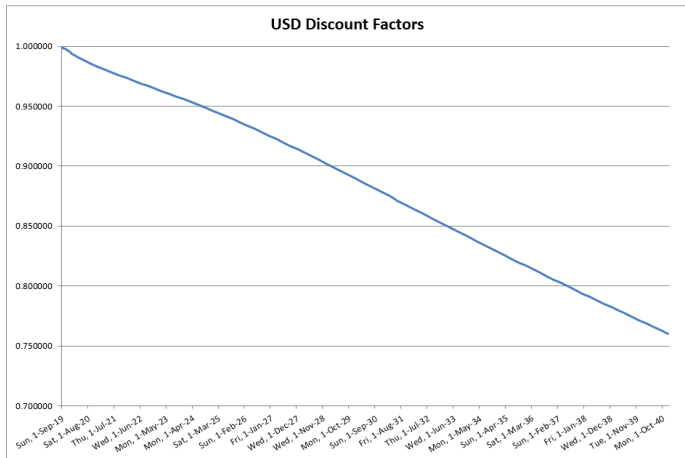


Discount Factors

Curve Collection	USDYC
CurveIndex	USDOIS

meCurveDiscountFactorsDisplay

PaymentDates	DiscountFactors
Mon, 2-Sep-19	1.000000
Tue, 3-Sep-19	0.999941
Wed, 4-Sep-19	0.999882
Wed, 11-Sep-19	0.999470
Wed, 18-Sep-19	0.999058
Wed, 25-Sep-19	0.998691
Fri, 4-Oct-19	0.998226
Mon, 4-Nov-19	0.996659
Wed, 4-Dec-19	0.995243
Mon, 6-Jan-20	0.993792
Tue, 4-Feb-20	0.992576
Wed, 4-Mar-20	0.991447
Thu, 4-Jun-20	0.988116
Fri, 4-Sep-20	0.985152
Fri, 4-Dec-20	0.982540
Thu, 4-Mar-21	0.980179
Fri, 4-Jun-21	0.977910
Tue, 7-Sep-21	0.975635
Mon, 6-Dec-21	0.973480
Fri, 4-Mar-22	0.971346
Mon, 6-Jun-22	0.969040
Tue, 6-Sep-22	0.966782
Mon, 5-Dec-22	0.964605
Mon, 6-Mar-23	0.962431
Mon, 5-Jun-23	0.960265
Tue, 5-Sep-23	0.958063
Mon, 4-Dec-23	0.955877
Mon, 4-Mar-24	0.953624
Tue, 4-Jun-24	0.951299
Wed, 4-Sep-24	0.948922
Wed, 4-Dec-24	0.946517
Tue, 4-Mar-25	0.944086



SWAPS: USD3ML

Swap Template

EffectiveDate	Maturity	Generator	PayRec	Notional	FixedRate	Spread
2D	3Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	4Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	5Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	7Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	8Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	9Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	10Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	12Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	15Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000
2D	20Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000

Trade Booking & Pricing

CreateSwap	CurveCollection	ParRate
USD_SWAP_3Y@9:4	USDYC	1.4000%
USD_SWAP_4Y@10:3	USDYC	1.3420%
USD_SWAP_5Y@11:3	USDYC	1.3182%
USD_SWAP_7Y@12:3	USDYC	1.3300%
USD_SWAP_8Y@13:3	USDYC	1.3430%
USD_SWAP_9Y@14:3	USDYC	1.3630%
USD_SWAP_10Y@15:3	USDYC	1.3892%
USD_SWAP_12Y@16:3	USDYC	1.4281%
USD_SWAP_15Y@17:3	USDYC	1.4751%
USD_SWAP_20Y@18:3	USDYC	1.5195%

meLWOSwapCreate

LEGTYPE	Leg1:Fixed	Leg2:Float
FORECASTCURVE		USD3ML
COMPOUNDMETHOD	NONE	NONE
DISCOUNTCURVE	USDOIS	USDOIS
VALUATIONCURRENCY	USD	USD
CURRENCY	USD	USD
FIRSTFIXING		
LASTFIXING		
FIRSTSTUBCURVEINDEX		NATURAL
LASTSTUBCURVEINDEX		NATURAL
NOTIONAL	1000000	1000000
NOTIONALEXCHANGE	NONE	NONE
LEVERAGE	1	1
PAYRECEIVE	PAY	RECEIVE
EFFECTIVEDATE	43712	43712
MATURITYDATE	44808	44808
ACCRUALFREQUENCY	Semi-Annual	Quarterly
ACCRUALBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
ACCRUALCALENDAR	NYB:LNb	NYB:LNb
ACCRUALDAYCOUNT	30/360	ACT/360
PAYMENTFREQUENCY	Semi-Annual	Quarterly
PAYMENTBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
PAYMENTCALENDAR	NYB:LNb	NYB:LNb
PAYMENTLAG	0D	0D
FIRSTSTUBDATE		
LASTSTUBDATE		
ROLLDAY	NORMAL	NORMAL
STUBTYPE	ShortStart	ShortStart
FIXINGBUSINESSDAYADJUSTMENT		MOD_FOLLOWING
FIXINGCALENDAR		NYB:LNb
FIXINGLAG		0D
FIXINGADVANCEORARREAR		
FIXEDRATE	1.0000%	
FLOATSPREAD		0

CreateSwap

Handle MYSWAP
CreateSwap MYSWAP@19:0
CurveCollection USDYC

Pricing & Risk

PV 11,763
PV01 294
ParRate 1.4000%
Annuity 2,940,677

Transparent Analytics

- Trade Bookings Easy to Display
- Calculation Breakdowns Highly Visible

DisplayCashflows

Leg1:Fixed Leg Sche

AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FixedRate	Coupon	DiscountFactor	CouponPV
04-Sep-19	04-Mar-20	182	0.5000	04-Mar-20	-1,000,000	0	1.00	1.0000%	-5,000	0.991447	-4,957
04-Mar-20	04-Sep-20	184	0.5000	04-Sep-20	-1,000,000	0	1.00	1.0000%	-5,000	0.985152	-4,926
04-Sep-20	04-Mar-21	181	0.5000	04-Mar-21	-1,000,000	0	1.00	1.0000%	-5,000	0.980179	-4,901
04-Mar-21	07-Sep-21	187	0.5083	07-Sep-21	-1,000,000	0	1.00	1.0000%	-5,083	0.975635	-4,959
07-Sep-21	04-Mar-22	178	0.4917	04-Mar-22	-1,000,000	0	1.00	1.0000%	-4,917	0.971346	-4,776
04-Mar-22	06-Sep-22	186	0.5056	06-Sep-22	-1,000,000	0	1.00	1.0000%	-5,056	0.966782	-4,888

Trade MYSWAP@164:1
CurveCollection USDYC

Bucketed Risk

USD : Pillars	USD : Total USD : Pillars		USD : Total
USDYC.OIS.OUTRIGHTOISSWAP.ON	0	USDYC.STD.O_N	0
USDYC.OIS.OUTRIGHTOISSWAP.1W	0	USDYC.STD.T_N	0
USDYC.OIS.OUTRIGHTOISSWAP.2W	0	USDYC.STD.LIBOR.3M	0
USDYC.OIS.OUTRIGHTOISSWAP.3W	0	USDYC.STD.LIBOR.6M	0
USDYC.OIS.OUTRIGHTOISSWAP.1M	0	USDYC.STD.FUTURE.ED1 COMDTY.20190918.20191218	0
USDYC.OIS.OUTRIGHTOISSWAP.2M	0	USDYC.STD.FUTURE.ED2 COMDTY.20191218.20200318	0
USDYC.OIS.OUTRIGHTOISSWAP.3M	0	USDYC.STD.FUTURE.ED3 COMDTY.20200318.20200617	0
USDYC.OIS.OUTRIGHTOISSWAP.4M	0	USDYC.STD.FUTURE.ED4 COMDTY.20200617.20200916	0
USDYC.OIS.OUTRIGHTOISSWAP.5M	0	USDYC.STD.FUTURE.ED5 COMDTY.20200916.20201216	0
USDYC.OIS.OUTRIGHTOISSWAP.6M	0	USDYC.STD.FUTURE.ED6 COMDTY.20201216.20210317	0
USDYC.OIS.OUTRIGHTOISSWAP.9M	0	USDYC.STD.FUTURE.ED7 COMDTY.20210317.20210616	0
USDYC.OIS.BASISRATE.1Y	0	USDYC.STD.FUTURE.ED8 COMDTY.20210616.20210915	0
USDYC.OIS.3MLIBORSWAP.1Y	0	USDYC.STD.3MLIBORSWAP.3Y	294
USDYC.OIS.BASISRATE.2Y	1	USDYC.STD.3MLIBORSWAP.4Y	0
USDYC.OIS.3MLIBORSWAP.2Y	-1	USDYC.STD.3MLIBORSWAP.5Y	0
USDYC.OIS.BASISRATE.3Y	1	USDYC.STD.3MLIBORSWAP.7Y	0
USDYC.OIS.3MLIBORSWAP.3Y	293	USDYC.STD.3MLIBORSWAP.8Y	0
USDYC.OIS.BASISRATE.4Y	0	USDYC.STD.3MLIBORSWAP.9Y	0
USDYC.OIS.3MLIBORSWAP.4Y	0	USDYC.STD.3MLIBORSWAP.10Y	0

GLOBAL CURVE ENGINE

Key	Value
CurveName	USD_CURVE
CurveCollection	USDYC
CurveObject	USD_CURVE@3:0
isCurveLoaded	TRUE

CURVE ENGINE PARAMETERS

FastRebuild	TRUE
Epsilon	1.00E-09
GradientEpsilon	1.00E-15
Delta	1.00E-10

CURVE GENERATORS

OIS	USD_OIS
3ML	USD_SWAP_3M

CURVE MARKET DATA

OIS	USD_OIS_MARKETDATA@1:0
3ML	USD_3M_MARKETDATA@2:0

MARKETDATA PROPERTIES

Key	Value
Currency	USD
CurveType	OIS
CurveIndexFrequency	DAILY
AsOfDate	Mon, 2-Sep-19

Create Market Data Handle

Handle Name	USD_OIS_MARKETDATA
Market Data Handle	USD_OIS_MARKETDATA@1:0

OIS

Term	Rate	StartDate	EndDate	Use
ON	1.91000%			TRUE
1W	1.91230%			TRUE
3M	2.04750%			TRUE
6M	2.16110%			TRUE

LIBOR BASIS SPREADS

Term	Rate
1Y	0.27500%
2Y	0.28750%
3Y	0.29688%
4Y	0.30625%
5Y	0.31250%
7Y	0.31813%
10Y	0.33750%
12Y	0.34630%
15Y	0.35875%
20Y	0.37625%
25Y	0.38500%
30Y	0.39500%
40Y	0.39500%
50Y	0.39500%

MARKETDATA PROPERTIES

Key	Value
Currency	USD
CurveType	SWAP
CurveIndexFrequency	QUARTERLY
AsOfDate	Mon, 2-Sep-19

Create Market Data Handle

Handle Name	USD_3M_MARKETDATA
Market Data Handle	USD_3M_MARKETDATA@2:0

LIBOR FIXINGS

Term	Rate
3M	2.3118%

FUTURES

Term	StartDate	EndDate	Rate	Volatility
ED1 Comdty	18-Dec-19	18-Mar-20	97.640	1.450%
ED2 Comdty	18-Mar-20	17-Jun-20	97.390	1.450%
ED3 Comdty	17-Jun-20	16-Sep-20	97.250	1.450%
ED4 Comdty	16-Sep-20	16-Dec-20	97.140	1.450%
ED5 Comdty	16-Dec-20	17-Mar-21	97.075	1.450%
ED6 Comdty	17-Mar-21	16-Jun-21	97.040	1.450%
ED7 Comdty	16-Jun-21	15-Sep-21	97.035	1.450%
ED8 Comdty	15-Sep-21	15-Dec-21	97.030	1.450%
ED9 Comdty	15-Dec-21	16-Mar-22	97.025	1.450%
ED10 Comdty	16-Mar-22	15-Jun-22	97.020	1.450%
ED11 Comdty	15-Jun-22	21-Sep-22	97.015	1.450%
ED12 Comdty	21-Sep-22	21-Dec-22	97.010	1.450%

SWAPS

Term	Rate	Use
4Y	2.8680%	TRUE
5Y	2.8729%	TRUE
7Y	2.8910%	TRUE
8Y	2.9014%	TRUE
9Y	2.9145%	TRUE
10Y	2.9280%	TRUE
12Y	2.9523%	TRUE
15Y	2.9728%	TRUE
20Y	2.9833%	TRUE
25Y	2.9750%	TRUE
30Y	2.9618%	TRUE
40Y	2.9618%	TRUE
50Y	2.9618%	TRUE

GLOBAL CURVE ENGINE

Key	Value
CurveName	EUR_CURVE
CurveCollection	EURYC
CurveObject	EUR_CURVE@6:0
IsCurveLoaded	TRUE

CURVE ENGINE PARAMETERS

FastRebuild	TRUE
Epsilon	1.00E-09
GradientEpsilon	1.00E-15
Delta	1.00E-10

CURVE GENERATORS

OIS	EUR_OIS
3ML	EUR_SWAP_3M

CURVE MARKET DATA

OIS	EUR_OIS_MARKETDATA@4:0
3ML	EUR_3M_MARKETDATA@5:0

MARKETDATAPROPERTIES

Key	Value
Currency	EUR
CurveType	OIS
CurveIndexFrequency	DAILY
AsOfDate	Mon, 2-Sep-19

Create Market Data Handle

Handle Name	EUR_OIS_MARKETDATA
Market Data Handle	EUR_OIS_MARKETDATA@4:0

OIS

Term	Rate	StartDate	EndDate	Use
ON	-0.36900%			TRUE
ECB1	-0.36500%	18-Sep-19	17-Dec-19	TRUE
ECB2	-0.36600%	18-Dec-19	24-Mar-20	TRUE
ECB3	-0.36900%	25-Mar-20	09-Jun-20	TRUE

LIBOROISBASISSPREADS

Term	Rate
1Y	0.05800%
2Y	0.06000%
3Y	0.07000%
4Y	0.08400%
5Y	0.09000%
7Y	0.09400%
10Y	0.09600%
12Y	0.09600%
15Y	0.09500%
20Y	0.09200%
25Y	0.08800%
30Y	0.08500%
40Y	0.07900%
50Y	0.07300%

MARKETDATAPROPERTIES

Key	Value
Currency	EUR
CurveType	SWAP
CurveIndexFrequency	QUARTERLY
AsOfDate	Mon, 2-Sep-19

Create Market Data Handle

Handle Name	EUR_3M_MARKETDATA
Market Data Handle	EUR_3M_MARKETDATA@5:0

LIBORFIXINGS

Term	Rate
3M	-0.3476%

FUTURES

Term	StartDate	EndDate	Rate	Volatility
ER1	18-Dec-19	18-Mar-20	100.310	0.400%
ER2	18-Mar-20	17-Jun-20	100.310	0.400%
ER3	17-Jun-20	16-Sep-20	100.305	0.400%
ER4	16-Sep-20	16-Dec-20	100.300	0.400%
ER5	16-Dec-20	17-Mar-21	100.300	0.400%
ER6	17-Mar-21	16-Jun-21	100.300	0.400%
ER7	16-Jun-21	15-Sep-21	100.300	0.400%
ER8	15-Sep-21	15-Dec-21	100.300	0.400%
ER9	15-Dec-21	16-Mar-22	100.275	0.400%
ER10	16-Mar-22	15-Jun-22	100.250	0.400%
ER11	15-Jun-22	21-Sep-22	100.180	0.400%
ER12	21-Sep-22	21-Dec-22	100.145	0.400%

SWAPS

Term	Rate	Use
4Y	-0.1750%	TRUE
5Y	-0.0650%	TRUE
6Y	0.0300%	TRUE
7Y	0.1280%	TRUE
8Y	0.2290%	TRUE
9Y	0.3260%	TRUE
10Y	0.4180%	TRUE
15Y	0.7760%	TRUE
20Y	0.9570%	TRUE
25Y	1.0270%	TRUE
30Y	1.0470%	TRUE
40Y	1.0450%	TRUE
50Y	1.0270%	TRUE

MARKETDATAPROPERTIES

Currency	EUR
CurveType	XCCYBASIS
CurveIndexFrequency	QUARTERLY
AsOfDate	Mon, 2-Sep-19

CREATE MARKET DATA HANDLE

Handle Name	EUR_XCCYBASIS_MARKETDATA
Market Data Handle	EUR_XCCYBASIS_MARKETDATA@7:0

CURVE DICTIONARY

HandleName	EUR_XCCY_CURVE
CurveGenerator	EUR_XCCY
DomesticCurve	EURYC
ForeignCurve	USDYC

CREATE CURVE HANDLE

Handle Name	EUR_XCCY_CURVE@8:0
Curve Collection	EURYC
Curve Index	XccyBasis:EURDF_USDCSA
Calibration Time Taken	0.04500 sec

FXSPOTS

Ccy1	Ccy2	Rate
EUR	USD	1.1202

FXFWDS

Term	Rate
ON	0.95
TN	0.98
1W	6.78
2W	13.42
3W	20.04
1M	30.36
2M	57.55
3M	86.39
4M	114.90
5M	143.35
6M	173.00
9M	259.48

XCCYSWAPS

Term	Rate
1Y	-0.14500%
2Y	-0.13500%
3Y	-0.13625%
4Y	-0.14250%
5Y	-0.14875%
6Y	-0.15500%
7Y	-0.16000%
8Y	-0.16125%
9Y	-0.16000%
10Y	-0.15875%
12Y	-0.15250%
15Y	-0.13375%
20Y	-0.08375%
25Y	-0.03875%
30Y	-0.00625%
40Y	-0.00625%
50Y	-0.00625%

XCCY: EURDF_USDCSA

Swap Template

EffectiveDate	Maturity	Generator	PayRec	Notional	Leg 1 Spread	Leg 2 Spread
2D	1Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	2Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	3Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	4Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	5Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	6Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	7Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	8Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	9Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	10Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	12Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	15Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	20Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	25Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000
2D	30Y	EUR_XCCY	PAY	1,000,000	0.00000%	0.0000

Trade Booking & Pricing

	CreateSwap	Spread
1	XCCYSWAP_1Y@9:0	-0.14500%
2	XCCYSWAP_2Y@10:0	-0.13500%
3	XCCYSWAP_3Y@11:0	-0.13625%
4	XCCYSWAP_4Y@12:0	-0.14250%
5	XCCYSWAP_5Y@13:0	-0.14875%
6	XCCYSWAP_6Y@14:0	-0.15500%
7	XCCYSWAP_7Y@15:0	-0.16000%
8	XCCYSWAP_8Y@16:0	-0.16125%
9	XCCYSWAP_9Y@17:0	-0.16000%
10	XCCYSWAP_10Y@18:0	-0.15875%
11	XCCYSWAP_12Y@19:0	-0.15250%
12	XCCYSWAP_15Y@20:0	-0.13375%
13	XCCYSWAP_20Y@21:0	-0.08375%
14	XCCYSWAP_25Y@22:0	-0.03875%
15	XCCYSWAP_30Y@23:0	-0.00625%

Curve Calibration Check

Term	Rate	+/- Bps
1Y	-0.14500%	0.00000
2Y	-0.13500%	0.00000
3Y	-0.13625%	0.00000
4Y	-0.14250%	0.00000
5Y	-0.14875%	0.00000
6Y	-0.15500%	0.00000
7Y	-0.16000%	0.00000
8Y	-0.16125%	0.00000
9Y	-0.16000%	0.00000
10Y	-0.15875%	0.00000
12Y	-0.15250%	0.00000
15Y	-0.13375%	0.00000
20Y	-0.08375%	0.00000
25Y	-0.03875%	0.00000
30Y	-0.00625%	0.00000

Pricing & Risk - Xccy Pricing Detail

Trade

XCXSWAP_3Y@11:0

SWAP_PROPERTIES

VALUATIONCURRENCY	USD
MTM	true
NOTIONALXRESETLEG	Leg2.Float

SWAP_LEG_PROPERTIES

LEGTYPE	Leg1.Float	Leg2.Float
FORECASTCURVE	EUR3ML	USD3ML
COMPOUNDMETHOD		
DISCOUNTCURVE	EURDF_USDCSA	USDOIS
VALUATIONCURRENCY	EUR	USD
CURRENCY	EUR	USD
FIRSTFIXING		
LASTFIXING		
FIRSTSTUBCURVEINDEX	NATURAL	NATURAL
LASTSTUBCURVEINDEX	NATURAL	NATURAL
ISPWDINTER		
NOTIONAL	1000000	1000000
NOTIONALEXCHANGE	START_AND_END	START_AND_END
REVERSE	1	1
PAYRECEIVE	PAY	RECEIVE
EFFECTIVEDATE	43712	43712
MATURITYDATE	44808	44808
ACCRUALFREQUENCY	QUARTERLY	QUARTERLY
ACCRUALBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
ACCRUALCALENDAR	UNB.NYB.TGT	UNB.NYB.TGT
ACCRUALDAYCOUNT	ACT/360	ACT/360
PAYMENTFREQUENCY		
PAYMENTBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
PAYMENTCALENDAR	UNB.NYB.TGT	UNB.NYB.TGT
PAYMENTLAG	00	00
FIRSTSTUBDATE		
LASTSTUBDATE		
STUBTYPE	NORMAL	NORMAL
FIXINGBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
FIXINGCALENDAR	UNB.NYB.TGT	UNB.NYB.TGT
FIXINGLAG	00	00
FIXINGADVANCEORARRAR		
FLOATSPREAD	0	0
CALCULATIONTYPE		
EXDIVIDENDTENOR		
EXDIVIDENDBUSINESSDAYADJUSTMENT		

ValuationSettings

Leg1.Float	EURYC
Leg2.Float	USDYC
FXSPOT	1.12020 EUR/USD

Display Cashflows

[eq1.FLOAT]eq1															
FixingDate	AccrualStart	AccrualEnd	AccrualDt	AccrualYe	PaymentDate	Notional	NotionalExch	Leverage	FloatSpre	FloatRate	Coupon	DiscountF	CouponPV		
04-Sep-19															
04-Sep-19	04-Sep-19	04-Dec-19	91	0.2528	04-Dec-19	-1,000,000	0	1.00	0.00	-0.3476%	879	1.000041	1,120,246		
04-Dec-19	04-Dec-19	04-Mar-20	91	0.2528	04-Mar-20	-1,000,000	0	1.00	0.00	-0.3159%	798	1.001936	986		
04-Mar-20	04-Mar-20	04-Jun-20	92	0.2556	04-Jun-20	-1,000,000	0	1.00	0.00	-0.3103%	793	1.005718	895		
04-Jun-20	04-Jun-20	04-Sep-20	92	0.2556	04-Sep-20	-1,000,000	0	1.00	0.00	-0.3063%	783	1.007586	884		
04-Sep-20	04-Sep-20	04-Dec-20	91	0.2528	04-Dec-20	-1,000,000	0	1.00	0.00	-0.3016%	762	1.009406	862		
04-Dec-20	04-Dec-20	04-Mar-21	90	0.2500	04-Mar-21	-1,000,000	0	1.00	0.00	-0.3015%	754	1.011209	854		
04-Mar-21	04-Mar-21	04-Jun-21	92	0.2556	04-Jun-21	-1,000,000	0	1.00	0.00	-0.3020%	772	1.013077	876		
04-Jun-21	04-Jun-21	07-Sep-21	95	0.2639	07-Sep-21	-1,000,000	0	1.00	0.00	-0.3027%	799	1.015082	908		
07-Sep-21	07-Sep-21	06-Dec-21	90	0.2500	06-Dec-21	-1,000,000	0	1.00	0.00	-0.3054%	799	1.017058	884		
06-Dec-21	06-Dec-21	04-Mar-22	88	0.2444	04-Mar-22	-1,000,000	0	1.00	0.00	-0.2817%	689	1.018969	786		
04-Mar-22	04-Mar-22	06-Jun-22	94	0.2611	06-Jun-22	-1,000,000	0	1.00	0.00	-0.2583%	674	1.020856	771		
06-Jun-22	06-Jun-22	06-Sep-22	92	0.2556	06-Sep-22	-1,000,000	-1,000,000	1.00	0.00	-0.1930%	-999,507	1.022414	-1,144,744		
[eq2.FLOAT]eq2															
FixingDate	AccrualStart	AccrualEnd	AccrualDt	AccrualYe	PaymentDate	Notional	NotionalExch	Leverage	FloatSpre	FloatRate	FxFixingDate	FxRate	Coupon	DiscountF	CouponPV
04-Sep-19															
04-Sep-19	04-Sep-19	04-Dec-19	91	0.2528	04-Dec-19	1,120,365	-7,947	1.00	0.00	2.3118%	04-Sep-19	0.89257	-1,400	0.994732	-1.393
04-Dec-19	04-Dec-19	04-Mar-20	91	0.2528	04-Mar-20	1,128,512	-8,645	1.00	0.00	2.3524%	04-Dec-19	0.88828	-1,936	0.989030	-1.915
04-Mar-20	04-Mar-20	04-Jun-20	92	0.2556	04-Jun-20	1,136,957	-9,673	1.00	0.00	2.5680%	04-Mar-20	0.87954	-1,211	0.983394	-1.191
04-Jun-20	04-Jun-20	04-Sep-20	92	0.2556	04-Sep-20	1,145,630	-8,811	1.00	0.00	2.7225%	04-Jun-20	0.87288	-840	0.977701	-822
04-Sep-20	04-Sep-20	04-Dec-20	91	0.2528	04-Dec-20	1,154,441	-9,250	1.00	0.00	2.8520%	04-Sep-20	0.86622	-985	0.971681	-957
04-Dec-20	04-Dec-20	04-Mar-21	90	0.2500	04-Mar-21	1,163,691	-9,633	1.00	0.00	2.8971%	04-Dec-20	0.85993	-1,205	0.965421	-1.163
04-Mar-21	04-Mar-21	04-Jun-21	92	0.2556	04-Jun-21	1,173,325	-10,188	1.00	0.00	2.9287%	04-Mar-21	0.85228	-1,406	0.958882	-1.348
04-Jun-21	04-Jun-21	07-Sep-21	95	0.2639	07-Sep-21	1,183,513	-10,706	1.00	0.00	2.9297%	04-Jun-21	0.84494	-1,556	0.952166	-1.482
07-Sep-21	07-Sep-21	06-Dec-21	90	0.2500	06-Dec-21	1,194,219	-10,165	1.00	0.00	2.9245%	07-Sep-21	0.83737	-1,434	0.945967	-1.357
06-Dec-21	06-Dec-21	04-Mar-22	88	0.2444	04-Mar-22	1,204,384	-9,889	1.00	0.00	2.9128%	06-Dec-21	0.83020	-1,295	0.940028	-1.217
04-Mar-22	04-Mar-22	06-Jun-22	94	0.2611	06-Jun-22	1,214,272	-10,444	1.00	0.00	2.9126%	04-Mar-22	0.82354	-1,210	0.933737	-1.130
06-Jun-22	06-Jun-22	06-Sep-22	92	0.2556	06-Sep-22	1,224,716	1,224,716	1.00	0.00	2.9032%	06-Jun-22	0.81652	1,233,803	0.927557	1,144,423

The End